2026 Housing EIS — Development Forecast

Project: 2026 Housing Element Environmental Impact Study (EIS)
Purpose: Parcel-level development forecasting for 2035 and 2050 model years to support the Tahoe Regional Transportation Plan (RTP) and Housing EIS alternatives analysis.

Overview

This folder contains the full forecasting workflow for the Lake Tahoe Basin Housing EIS. The workflow is split into two phases: **shared data engineering** (parcel preparation and base unit pool assembly) and **scenario-specific forecasting** (per-alternative unit allocation, TAZ summaries, and travel demand model inputs). Four housing policy alternatives are evaluated, each with its own unit distribution assumptions and known-projects list.

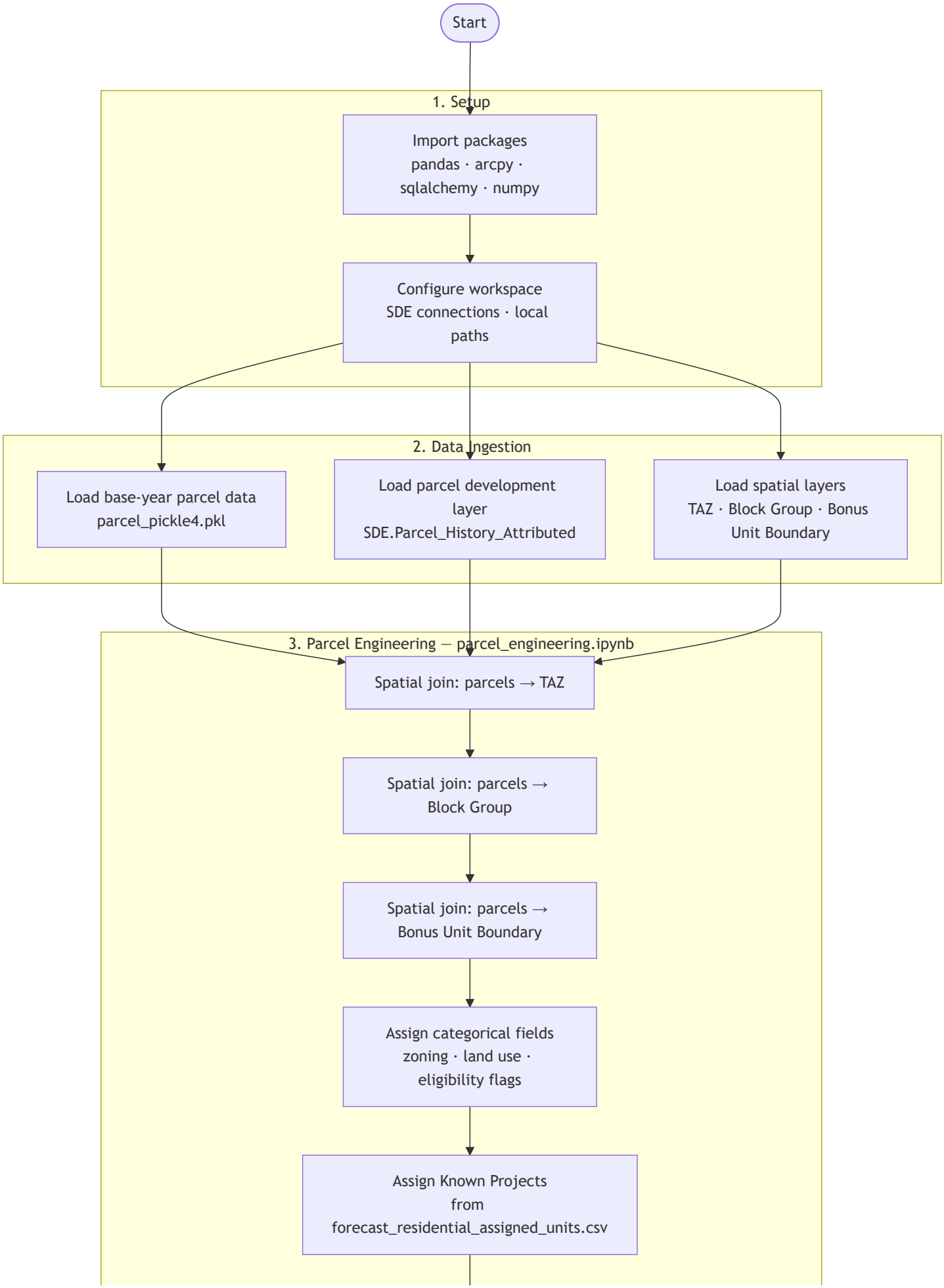
Folder Structure

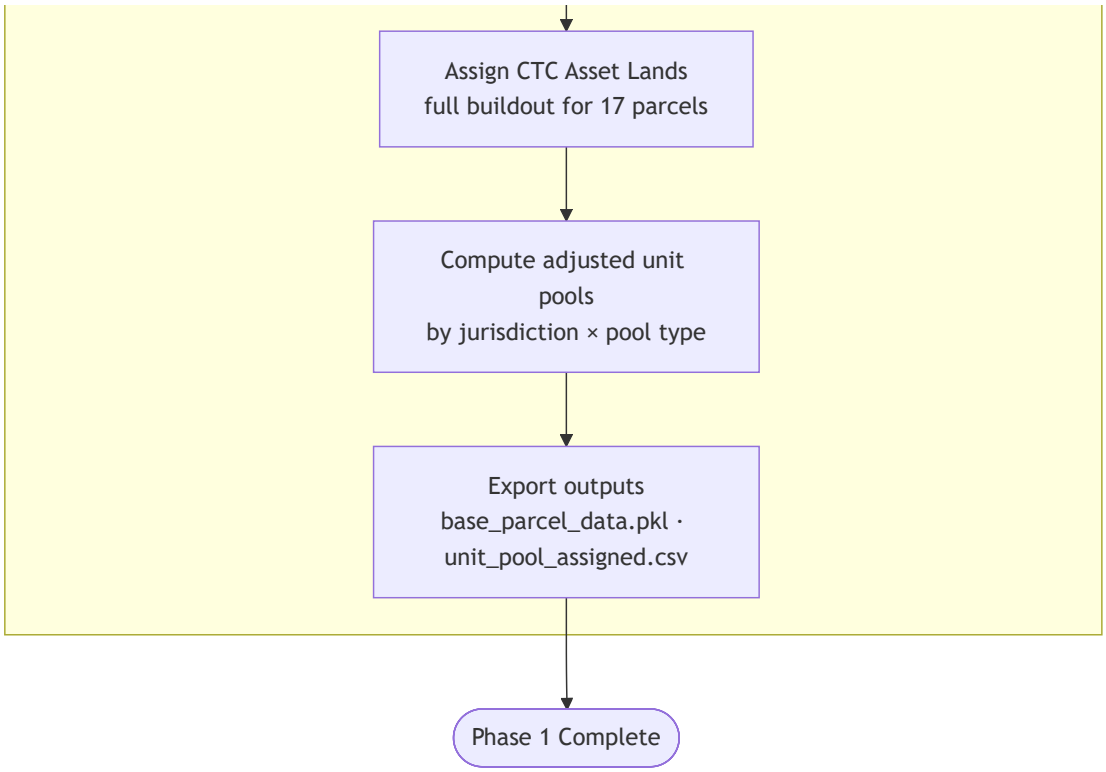
```
Forecast/
├── README.md                          # This file
├── scripts/                           # Shared Phase 1 scripts (run once, common to all alterna
│   ├── parcel_engineering.ipynb       # Phase 1A: spatial joins, parcel classification, unit po
│   ├── scenario_forecast_template.ipynb  # Phase 1B: reference template for scenario notebooks
│   ├── Base_Forecast.ipynb            # Original combined notebook (source of Phase 1A/1B)
│   ├── utils.py                       # Shared helper functions
│   └── Lookup_Lists/                  # Shared input CSVs (zoned units, assigned units, CTC lan
├── data/                              # Shared processed data outputs
│   ├── base_parcel_data.pkl           # Engineered parcel layer (output of Phase 1A)
│   ├── unit_pool_assigned.csv         # Adjusted unit pools after known-project subtraction
│   ├── taz_summary_2035.csv/.pickle   # TAZ-level summary — 2035 model year
│   ├── taz_summary_2050.csv/.pickle   # TAZ-level summary — 2050 model year
│   ├── Parcels_Forecast.csv           # Parcel-level forecast (tabular)
│   └── TAZ_Units.csv                  # Existing and forecasted residential units by TAZ
├── Alternative_1/                     # Housing Alternative 1 scenario
│   ├── Alternative_1_Forecast.ipynb
│   ├── utils.py
│   ├── inputs/                        # Alternative-specific zoned units and scenario distribut
│   └── Changes from RTP Model Run.md
├── Alternative_2/                     # Housing Alternative 2 scenario
│   ├── Alternative_2_Forecast.ipynb
│   └── inputs/
├── Alternative_3/                     # Housing Alternative 3 scenario
│   ├── Alternative_3_Forecast.ipynb
│   └── inputs/
└── Alternative_4/                     # Housing Alternative 4 scenario
    ├── Alternative_4_Forecast.ipynb
    └── inputs/
```

Process Flow

Phase 1 — Shared Data Engineering (`scripts/`)

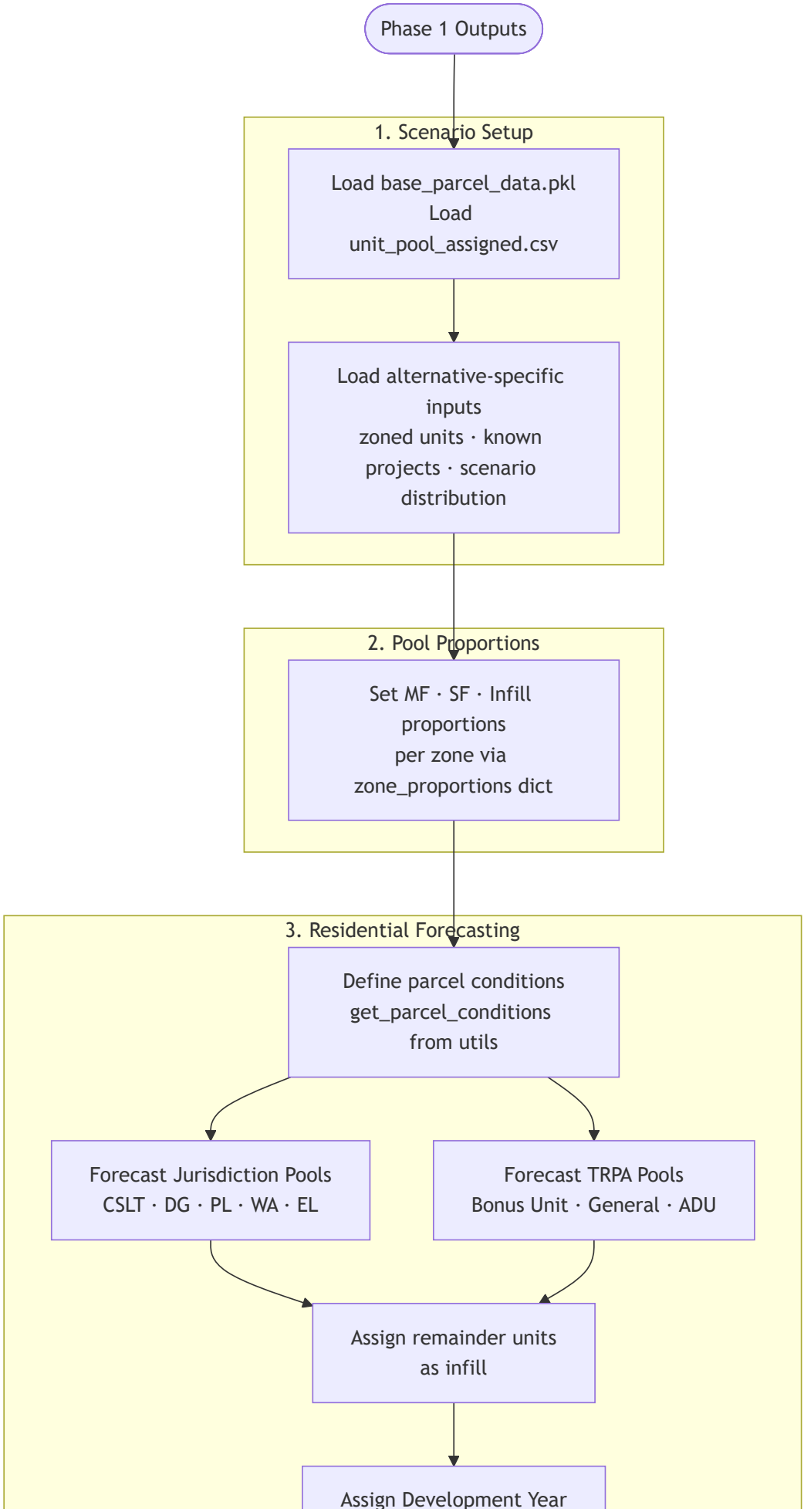
Run once. Outputs are shared across all alternatives.

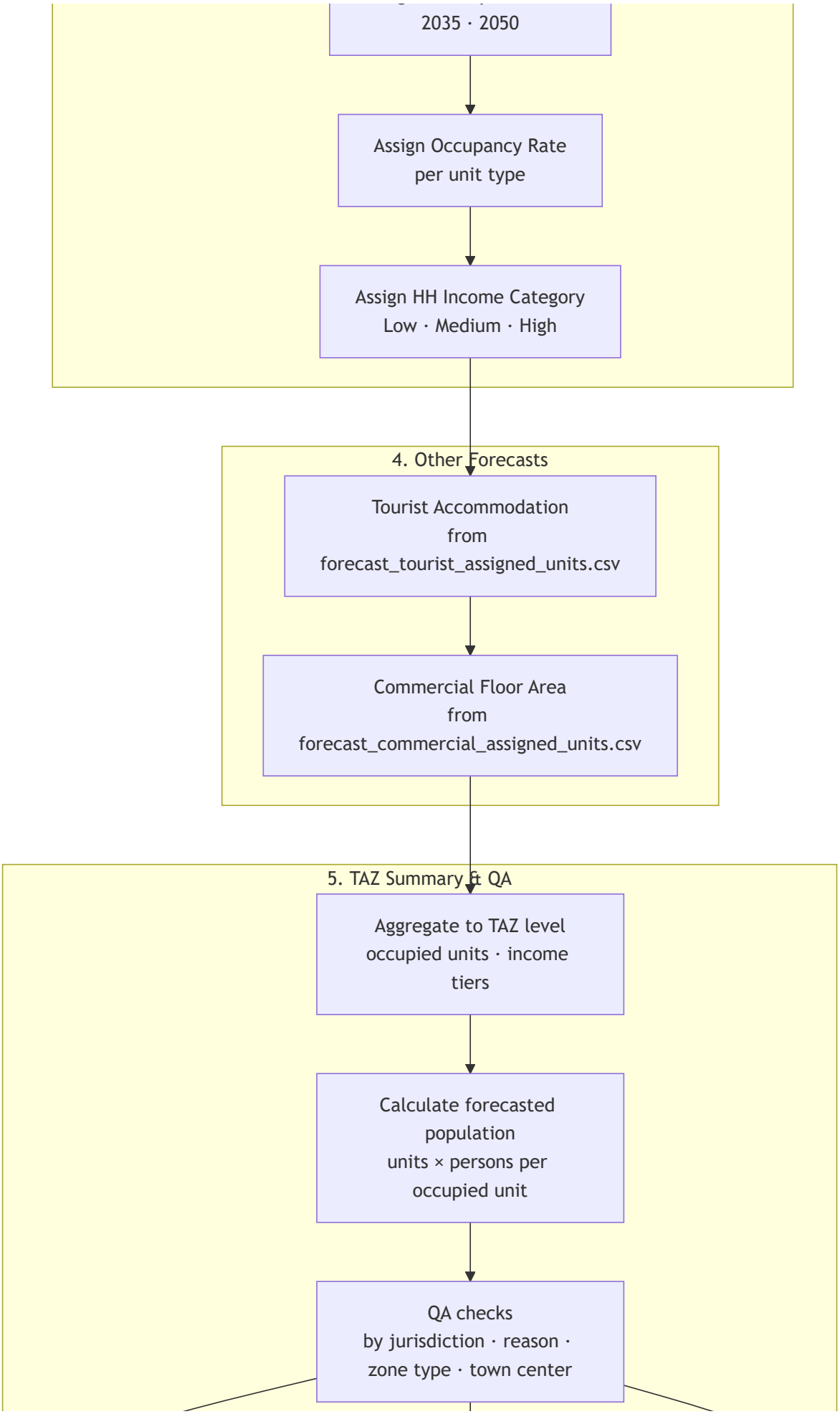

Start
1. Setup
Import packages
pandas · arcpy ·
sqlalchemy · numpy
Configure workspace
SDE connections · local
paths
2. Data Ingestion
Load base-year parcel data
parcel_pickle4.pkl
Load parcel development
layer
SDE.Parcel_History_Attributed
Load spatial layers
TAZ · Block Group · Bonus
Unit Boundary
3. Parcel Engineering — parcel_engineering.ipynb
Spatial join: parcels → TAZ
Spatial join: parcels →
Block Group
Spatial join: parcels →
Bonus Unit Boundary
Assign categorical fields
zoning · land use ·
eligibility flags
Assign Known Projects
from
forecast_residential_assigned_units.csv

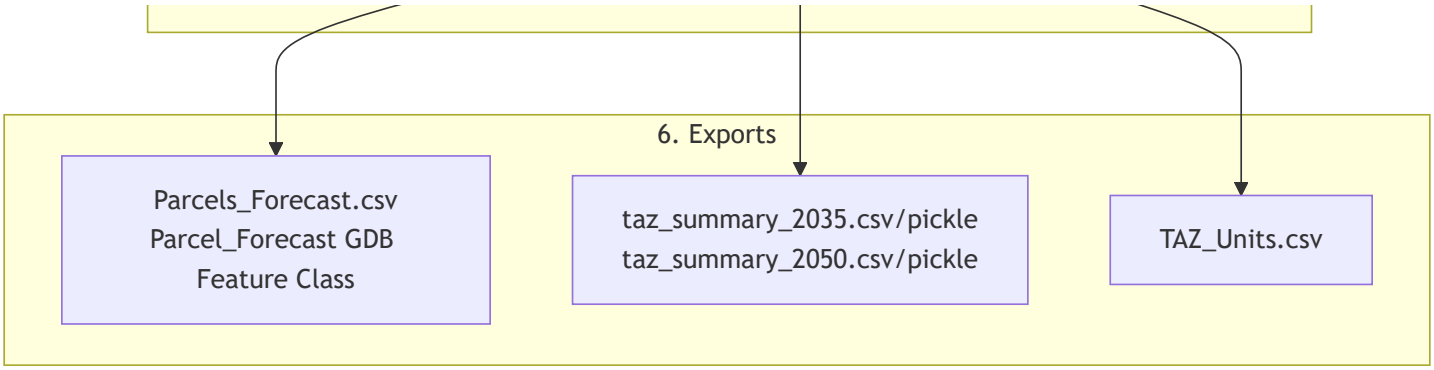


Phase 2 — Scenario Forecasting (per Alternative)

Run independently for each alternative using its own `inputs/` folder.


Phase 1 Outputs
1. Scenario Setup
Load base_parcel_data.pkl
Load
unit_pool_assigned.csv
Load alternative-specific
inputs
zoned units · known
projects · scenario
distribution
2. Pool Proportions
Set MF · SF · Infill
proportions
per zone via
zone_proportions dict
3. Residential Forecasting
Define parcel conditions
get_parcel_conditions
from utils
Forecast Jurisdiction Pools
CSLT · DG · PL · WA · EL
Forecast TRPA Pools
Bonus Unit · General · ADU
Assign remainder units
as infill
Assign Development Year


2035 · 2050
Assign Occupancy Rate per unit type
Assign HH Income Category Low · Medium · High
4. Other Forecasts
Tourist Accommodation from forecast_tourist_assigned_units.csv
Commercial Floor Area from forecast_commercial_assigned_units.csv
5. TAZ Summary & QA
Aggregate to TAZ level occupied units · income tiers
Calculate forecasted population units × persons per occupied unit
QA checks by jurisdiction · reason · zone type · town center



Notebooks

Notebook	Phase	Purpose
`scripts/parcel_engineering.ipynb`	1A	Spatial joins, parcel classification, unit pool assembly; exports `base_parcel_data.pkl` and `unit_pool_assigned.csv`
`scripts/scenario_forecast_template.ipynb`	1B	Reference template for building alternative notebooks; loads Phase 1A outputs
`scripts/Base_Forecast.ipynb`	—	Original combined notebook (source for the Phase 1A/1B split); kept for reference
`Alternative_1/Alternative_1_Forecast.ipynb`	2	Scenario forecast for Alternative 1
`Alternative_2/Alternative_2_Forecast.ipynb`	2	Scenario forecast for Alternative 2
`Alternative_3/Alternative_3_Forecast.ipynb`	2	Scenario forecast for Alternative 3
`Alternative_4/Alternative_4_Forecast.ipynb`	2	Scenario forecast for Alternative 4

Inputs

Shared Inputs (`scripts/Lookup_Lists/`)

Input	Type	Description
`forecast_residential_assigned_units.csv`	CSV	Known residential allocations 2023–2025
`forecast_residential_zoned_units.csv`	CSV	Jurisdiction-level zoned residential unit pools
`forecast_tourist_assigned_units.csv`	CSV	Known tourist accommodation changes
`forecast_tourist_zoned_units.csv`	CSV	Jurisdiction-level zoned tourist unit pools
`forecast_commercial_assigned_units.csv`	CSV	Known commercial floor area changes
`forecast_commercial_zoned_units.csv`	CSV	Jurisdiction-level zoned commercial floor area
`CTC_AssetLands_Lookup.csv`	CSV	17 CTC asset land parcels for full buildout
`known_projects.csv`	CSV	Master list of known development projects
`SocioEcon_Summer.csv`	CSV	Base-year TAZ socioeconomic data (persons per unit)

Alternative-Specific Inputs (`Alternative_N/inputs/`)

Input	Type	Description
`forecast_residential_zoned_units.csv`	CSV	Scenario-adjusted zoned unit pool overrides
`known_projects.csv`	CSV	Alternative-specific known projects list
`scenarioN_instructions.csv`	CSV	Narrative description of scenario assumptions
`scenarioN_unit_distribution.csv`	CSV	Per-zone MF / SF / Infill proportion targets

Spatial / Database Inputs

Input	Type	Description
`parcel_pickle4.pkl`	Pickle	Base-year parcel data from the 2022 Travel Demand Model
`SDE.Parcel_History_Attributed`	SDE Feature Class	Parcel development records with attributed history
TAZ polygons	SDE Feature Class	Traffic Analysis Zones
Block Group polygons	SDE Feature Class	Census block groups
Bonus Unit Boundary	SDE Feature Class	TRPA bonus unit eligibility boundary

Outputs

Shared (`data/`)

Output	Type	Description
`base_parcel_data.pkl`	Pickle	Engineered parcel layer with spatial joins and classification (Phase 1A output)
`unit_pool_assigned.csv`	CSV	Unit pools after subtracting known projects and CTC lands
`Parcels_Forecast.csv`	CSV	Parcel-level forecast with all assigned attributes
`taz_summary_2035.pkl/.csv`	Pickle / CSV	TAZ-level summary for 2035 model year
`taz_summary_2050.pkl/.csv`	Pickle / CSV	TAZ-level summary for 2050 model year
`TAZ_Units.csv`	CSV	Existing and forecasted residential units by TAZ

Per Alternative

Output	Type	Description
`Parcel_Forecast`	GDB Feature Class	Spatial parcel forecast for GIS use
`taz_summary_2035.csv/.pickle`	CSV / Pickle	Alternative-specific TAZ summary — 2035
`taz_summary_2050.csv/.pickle`	CSV / Pickle	Alternative-specific TAZ summary — 2050

Key Logic

Two-Phase Architecture

The workflow is intentionally split so that the computationally expensive parcel engineering (spatial joins, SDE reads) only runs once. All four alternatives share the same engineered parcel layer and start from the same adjusted unit pools.

Phase	Script	Runs	Key Output
1A — Parcel Engineering	`parcel_engineering.ipynb`	Once	`base_parcel_data.pkl` , `unit_pool_assigned.csv`
1B — Scenario Template	`scenario_forecast_template.ipynb`	Reference only	—
2 — Scenario Forecast	`Alternative_N_Forecast.ipynb`	Once per alternative	TAZ summaries, parcel forecast

Residential Unit Allocation

Units are allocated in a strict priority order:

1. **Known Projects** — directly mapped by APN from the assigned units lookup

2. **CTC Asset Lands** — full buildout assigned to 17 specific parcels
3. **Jurisdictional Pools** — remaining units distributed by jurisdiction (CSLT, DG, PL, WA, EL) and unit type via the `zone_proportions` dictionary:
   - Multifamily (MF) — default 35% of pool (overridable per zone)
   - Single-family (SF) — default 50% of pool (overridable per zone)
   - Infill — default 15% of pool (overridable per zone)
4. **TRPA Pools** — bonus unit and general allocations administered by TRPA
5. **Remainders** — leftover pool units assigned as infill
6. **ADUs** — Accessory Dwelling Units filled to reach regional total

Pool Proportion Overrides

Each alternative notebook contains a `zone_proportions` dictionary keyed by `(Jurisdiction, Unit_Pool)`. Any zone not listed falls back to global defaults. This allows fine-grained scenario differentiation without changing core allocation logic.

```
zone_proportions = {
    ('CSLT', 'Bonus Unit'): {'mf': 0.50, 'sf': 0.35, 'infill': 0.15},
    # ... more overrides ...
    'default': {'mf': 0.35, 'sf': 0.50, 'infill': 0.15},
}
```

Development Year Assignment

Year	Share	Method
2035	~33% of total	All "Assigned" projects + proportional random draw
2050	Remainder	Remaining parcels

Parcel Eligibility Conditions

Parcels are filtered by jurisdiction-specific criteria before unit assignment (defined in `utils.get_parcel_conditions()`):

- **Vacant buildable:** private, not retired, IPES score > 0 (Placer: > 726)
- **Infill:** existing residential with specific zoning
- **Bonus unit boundary:** must be within the TRPA-designated polygon
- **ADU eligible:** existing single residential unit with ADU allowed flag set

Population Calibration

Forecasted population = occupied units × persons per occupied unit (from base-year socioeconomic data). The persons-per-unit factor is adjusted iteratively until regional population targets are met:

- **2035 target:** 55,592 persons
- **2050 target:** 57,611 persons

Key Changes from RTP Model Run

The following changes are common across all alternatives (documented in each `Changes from RTP Model Run.md`):

1. Known residential unit changes from 2022–2025 are pre-incorporated into the input parcel pickle
2. Adjusted zoned unit pools to account for construction between 2022 and 2025 (per K. Kasman)
3. Updated known projects list
4. ADUs are now allowed in Washoe County
5. 33% of allocations assumed to be multi-family (100% occupied); remaining 66% single-family
6. Population in 2035 and 2050 is allowed to vary based on persons per household and occupancy rate (not fixed to a target)

Dependencies

Package	Use
`pandas`	Data manipulation
`arcpy`	Spatial joins, SDE reads, feature class export
`numpy`	Numeric operations
`sqlalchemy`	SQL Server database connections
`pathlib`	File path management
`pickle`	Intermediate data serialization

Package	Use
`utils.py`	Project-specific helper functions (`get_parcel_conditions` , `forecast_residential_units` , `forecast_residential_units_infill` , `get_target_sum` , `check_parcel_condition` , etc.)

> **Environment:** Requires an ArcGIS Pro Python environment with access to the TRPA SDE geodatabase and SQL Server instances (`sql12` , `sql14`). Database credentials are read from the `DB_USER` and `DB_PASSWORD` environment variables.